

Chernobyl Chemical Radiation Analysis

Anthony
School of Computer Science
Bina Nusantara University
Jakarta Barat, Indonesia
anthony021@binus.ac.id

Hazel Zaki Adityo
School of Computer Science
Bina Nusantara University
Jakarta Barat, Indonesia
hazel.adityo@binus.ac.id

Satya Darma Padmakumara
School of Computer Science
Bina Nusantara University
Jakarta Barat, Indonesia
satya.padmakumara@binus.ac.id

Abstract— "The Chernobyl Nuclear Power Plant disaster that occurred in 1986 was a massive disaster that released significant quantities of radioactive isotopes, which requires long-term environmental analysis. This study aims to statistically model the concentration of Caesium-137 (Cs-137) based on its correlation with other radionuclides, specifically Caesium-134 (Cs-134) and Iodine-131 (I-131), and geographic coordinates. Using a dataset of 1,415 observations from European countries, this study applies Multiple Linear Regression to analyze the relationship between these variables. Four different models were evaluated: Basic Linear Model, Log Transform Model, Interaction Model, and Fixed Effects Model⁴. The results indicate that Cs-134 is the strongest predictor of Cs-137 levels, followed by I-131, reflecting their simultaneous release and chemical similarity. Although the Interaction Model achieved the highest statistical fit ($R^2 = 0.8047$), the Basic Linear Model was chosen as the final model due to its simplicity and interpretability, explaining 78.77% of the variance. Geographic analysis revealed a small negative correlation with latitude, while longitude showed no statistically significant effect on radiation levels. The results substantiate the strong correlation between radioactive isotope distributions and highlight the utility of regression analysis in environmental radiation modeling."

Keywords— Caesium-137, Chernobyl accident, environmental radiation, multiple linear regression, radioactive isotopes.

I. INTRODUCTION

The Chernobyl Nuclear Power Plant disaster, which occurred on April 26, 1986, in Pripyat, Ukraine, stands as one of the most catastrophic nuclear events in history. It was precipitated by a group of technicians with limited understanding of nuclear reactor physics conducting an electrical experiment at Reactor Unit 4. This explosion released a massive quantity of radioactive material into the atmosphere.

The aftermath had severe detrimental effects on human health. Three specific radionuclides are of primary concern: Iodine-131 (I-131), Caesium-134 (Cs-134), and Caesium-137 (Cs-137). Each of these substances poses distinct environmental impacts. These isotopes contribute to long-term health risks, disruption of food chains, as well as soil and atmospheric contamination.

Analyzing the measurement data of these radioactive substances remains crucial, despite the decades that have passed since the incident. The raw data exhibits significant variability, heavily dependent on geographic location and sampling timing.

One of the primary challenges in understanding the disaster's full impact lies in modeling the correlations between these radioactive variables. A statistical approach, specifically Regression Analysis, is required to elucidate the relationships between these variables.

This study focuses on modeling the linear correlation between the three radioactive isotopes—Iodine-131 (I-131), Caesium-134 (Cs-134), and Caesium-137 (Cs-137)—and analyzing their concentration levels based on geographic location. Through this research, it is expected that the variability of Chernobyl radiation data can be understood and explained, serving as a valuable reference for mitigating environmental impacts in future nuclear incidents.

II. THEORETICAL FRAMEWORK

A. Radioactive Decay Theory

The fundamental basis for changes in the concentration of radioactive substances in a dataset is explained by the Law of Exponential Decay. Unstable radionuclides decay into other isotopes over time, emitting radiation in the process. The relationship between the number of particles and time is described mathematically:

$$N(t) = N_0 e^{-\lambda t}$$

Where $N(t)$ is the number of particles at time t , N_0 is the initial number, and λ is the decay constant. This theory is the basis for why the Time variable has a significant effect on the decrease in the concentration of Iodine-131 (which has a short half-life of 8 days) compared to Caesium-137 (a long half-life of 30 years).

B. Geochemical Characteristics of Radionuclides

This theoretical framework should also address the specific chemical properties of the variables you are studying:

1. **Iodin-131 (I-131):** Volatile (easily evaporates). Highly reactive and spreads rapidly through the air immediately after the explosion.
2. **Caesium-137 (Cs-137) & Cs-134:** Strongly bound to aerosol particles. Tends to settle to the ground through rain and persists for long periods in the topsoil. These differences in physicochemical properties underlie the hypothesis that the correlation between iodine and cesium may not always be perfectly linear due to the different transport mechanisms in nature.

III. METHODOLOGY

A. Dataset Overview

For this study, we utilized the “Chernobyl Chemical Radiation” dataset, which is publicly available on Kaggle. This dataset aggregates chemical measurements and radiation levels recorded in the wake of the Chernobyl nuclear disaster, covering a wide range of geographic locations and isotope concentrations. While this data is often used for simple geospatial mapping, our research takes a different approach by applying inferential statistics through Multiple Linear Regression. Our primary goal was to statistically model the behavior of Cesium-137 (Cs-137) by analyzing its relationship with other isotopes (Cs-134, I-131) and specific geographic coordinates.

B. Variable Used

We narrowed down the dataset's original variables to focus specifically on quantitative metrics relevant to our regression goals. We prioritized variables that provided direct measurements of radioactive isotopes and precise geographic positioning. To ensure the integrity of our analysis, we performed rigorous data cleaning; specifically, we removed any observations that contained missing values for our target variables, ensuring a "complete cases" dataset for modeling. The specific variables analyzed are detailed below:

TABLE I. VARIABLES USED IN THE RESEARCH

Variable Name	Description
Cs_137	Concentration of Cesium-137 isotope (Dependent Variable (Y))
Cs_134	Concentration of Cesium-134 isotope (Predictor Variable (X1))
I_131	Concentration of Iodine-131 isotope (Predictor Variable (X2))
Latitude	Geographic latitude coordinate (Predictor Variable (X3))
Longitude	Geographic longitude coordinate (Predictor Variable (X4))

Country	Used for the Fixed Effects model (Control Variable (Categorical))
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C. Research Steps

Our analysis pipeline was implemented in R and progressed through several systematic stages. First, we preprocessed the data by standardizing column names, converting all radiation measurements to numeric types, and filtering out incomplete rows. Second, we conducted an Exploratory Data Analysis (EDA), generating correlation and scatterplot matrices to visually inspect relationships between the isotopes and check for potential multicollinearity before building our models.

Third, we moved to the modeling phase. Using the `lm()` function for standard regression and `rlm()` from the MASS package for robust regression, we constructed and compared five distinct models to find the best fit:

1. **Basic Linear Model:** A standard approach proposing that Cs-137 is a function of Cs-134, I-131, and location.
2. **Log-Transformed Model:** We applied natural logarithms to the isotope data to handle potential skewness in the distribution.
3. **Interaction Model:** This model tested if the effect of one isotope depends on the level of another (interaction terms).
4. **Fixed Effects Model:** We added "Country" as a factor to control for regional differences.
5. **Robust Regression:** We used M-estimation to minimize the impact of any outliers in the radiation data.

Finally, we validated our assumptions using the Shapiro-Wilk test for normality, the Breusch-Pagan test (via `lmtest`) for homoscedasticity, and Variance Inflation Factors (via `car`) to ensure multicollinearity was not biasing our results.

D. Model Accuracy Metrics

To evaluate which model performed best, we looked beyond just one metric. We compared R-squared and Adjusted R-squared values to see how well each model explained the variance in the data. We also assessed the trade-off between model fit and complexity using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) where lower values signaled a more efficient model. Additionally, we calculated the Root Mean Squared Error (RMSE) to quantify the average prediction error.

E. Theoretical Foundation: Multiple Linear Regression

Our theoretical framework assumes a linear relationship between the concentration of Cs-137 and the presence of other isotopes likely due to their simultaneous release during the reactor explosion and subsequent decay chains. We also posited that geographic location acts as a proxy for the distance and direction of the fallout plume. By employing Multiple Linear Regression, we were able to isolate the specific impact of each predictor (such as Cs-134 levels or

Latitude) on Cs-137 concentrations while holding other factors constant (*ceteris paribus*).

IV. RESULTS AND DISCUSSION

F. Descriptive Statistics

This study utilizes secondary data from the Chernobyl Chemical Radiation dataset, consisting of radiation measurements collected across several European countries following the Chernobyl nuclear accident in 1986. After data cleaning and removal of incomplete observations, a total of 1,415 observations were used in the analysis. The dependent variable is Cesium-137 (Cs-137), while the independent variables include Cesium-134 (Cs-134), Iodine-131 (I-131), Latitude, and Longitude.

Preliminary analysis indicates that radiation variables are positively skewed, with most observations recording low radiation levels and a smaller number of observations exhibiting very high values. This pattern is typical for radioactive fallout data, where contamination is concentrated in specific regions rather than evenly distributed. Spatially, higher radiation concentrations tend to appear in Central and Eastern Europe, while lower concentrations are observed in Western and Northern Europe.

G. Correlation Analysis

Correlation analysis was conducted to examine the relationships among the variables prior to regression modeling. The results show a strong positive correlation between Cs-137 and Cs-134, as well as a strong positive correlation between Cs-137 and I-131. This indicates that areas with high Cs-134 and I-131 concentrations tend to also have high Cs-137 concentrations.

The correlation between Cs-134 and I-131 is moderate, suggesting that although both isotopes originate from the same nuclear event, their behavior differs due to distinct physical properties and decay rates. The correlations between Cs-137 and the geographic variables (Latitude and Longitude) are relatively weak, indicating that location alone does not strongly determine radiation levels without considering radioactive isotope concentrations.

These results support the inclusion of Cs-134 and I-131 as primary predictors in the regression model.

H. Basic Linear Regression

The first model estimated is the base multiple linear regression model with the following specification:

$$Cs_{137} = \beta_0 + \beta_1 Cs_{134} + \beta_2 I_{131} + \beta_3 Latitude + \beta_4 Longitude + \varepsilon$$

The regression results show that the model is statistically significant with an F-statistic p-value of less than 0.001. The model achieves an R^2 value of 0.7877, indicating that 78.77% of the variation in Cs-137 concentrations is explained by the independent variables.

Cs-134 has the strongest and most significant effect on Cs-137 ($\beta = 1.228$, $p < 0.001$). This means that for every one-unit increase in Cs-134, Cs-137 increases by approximately

1.228 units, holding other variables constant. This result is consistent with the physical nature of radioactive release, as both isotopes are cesium and were emitted simultaneously during the nuclear accident.

I-131 also shows a positive and statistically significant effect ($\beta = 0.123$, $p < 0.001$). Although its effect is smaller compared to Cs-134, this result indicates that I-131 contributes meaningfully to explaining Cs-137 concentrations.

Latitude has a small but statistically significant negative coefficient ($\beta = -0.003$, $p = 0.031$), suggesting that locations further north tend to record slightly lower Cs-137 concentrations. Longitude is not statistically significant ($p = 0.961$), indicating that east-west position does not contribute meaningfully to predicting Cs-137 once other variables are controlled.

I. Log-Transformed Regression

To address skewness in the data and potential non-normality of residuals, a log-transformed regression model was estimated. In this model, the natural logarithms of Cs-137, Cs-134, and I-131 were used, while Latitude and Longitude remained in their original scale.

The results indicate that both $\log(Cs-134)$ and $\log(I-131)$ are positive and statistically significant predictors ($p < 0.001$). Latitude also remains statistically significant with a negative coefficient, while Longitude remains insignificant.

However, this model produces an R^2 value of 0.5663, which is substantially lower than that of the base model. Although the log transformation improves certain distributional properties, the overall explanatory power is reduced. In addition, the interpretation of logarithmic coefficients is less intuitive. Therefore, this model is not selected as the main model.

J. Interaction Model

The third model incorporates interaction terms between Cs-134 and I-131, as well as between Latitude and Longitude, to examine whether the effect of one variable depends on the level of another.

The interaction between Cs-134 and I-131 is statistically significant ($p < 0.001$), indicating that the effect of Cs-134 on Cs-137 varies depending on the level of I-131. This suggests a synergistic relationship between the isotopes in influencing radiation levels. In contrast, the interaction between Latitude and Longitude is not statistically significant.

This model achieves the highest R^2 value of 0.8047, indicating that it provides the best statistical fit among all models. The lower RMSE and AIC values further support its strong performance. However, the inclusion of interaction terms increases model complexity and reduces interpretability.

K. Country Fixed Effects Model

The fourth model includes country dummy variables to control for unobserved country-specific factors such as environmental conditions, measurement practices, and policy differences. Several country indicators, including Germany, France, the Netherlands, Norway, Sweden, and the United Kingdom, are statistically significant.

The model achieves an R^2 value of 0.8045, which is slightly lower than that of the interaction model but higher than the base model. Cs-134 and I-131 remain highly significant, and Latitude becomes more strongly significant. Longitude remains statistically insignificant.

The inclusion of country fixed effects improves model fit but also substantially increases the number of parameters, making the model less parsimonious.

L. Model Comparison and Selection

TABLE II. MODEL COMPARISON TABLE

Model	R^2	Adj R^2	AIC	RMSE
Model 1	0.7877	0.7871	2002	0.489
Model 2	0.5663	0.5650	6128	2.100
Model 3	0.8047	0.8038	1889	0.469
Model 4	0.8045	0.8019	1914	0.469

A comparison of the four models was conducted using R^2 , adjusted R^2 , AIC, BIC, and RMSE. The interaction model (Model 3) demonstrates the best statistical performance with the highest R^2 (0.8047), lowest AIC (1889), and lowest RMSE (0.469). The fixed effects model (Model 4) shows comparable performance but with greater complexity.

Despite the superior statistical performance of Model 3, Model 1 is selected as the final model due to its balance between explanatory power, simplicity, and interpretability. Model 1 explains a substantial proportion of variance while remaining easy to interpret and align with the primary objective of the study.

M. Diagnostic Tests and Assumption Checking

Several diagnostic tests were conducted to evaluate whether the classical assumptions of multiple linear regression were satisfied.

The Shapiro-Wilk test indicates that the residuals are not normally distributed ($p < 0.001$). This is expected given the highly skewed nature of radiation data. However, given the large sample size, the violation of normality is considered

acceptable and does not significantly affect the reliability of coefficient estimates.

The Breusch-Pagan test indicates the presence of heteroscedasticity ($p < 0.001$), suggesting that the variance of residuals is not constant. This implies that prediction errors increase at higher radiation levels. Although this violates the homoscedasticity assumption, the model remains suitable for interpretation in an academic context.

Variance Inflation Factor (VIF) values for all independent variables are below 2, indicating that no multicollinearity problem exists. This means that the predictors are not highly correlated with each other and each contributes unique information to the model.

Visual inspection of the data suggests some degree of clustering due to repeated measurements at the same locations over time, which may slightly violate the independence assumption. Nevertheless, for the purpose of this study, the model is considered adequate.

N. Discussion

The results of this study demonstrate that Cs-134 and I-131 are the primary determinants of Cs-137 concentrations following the Chernobyl nuclear accident. The strong relationship between Cs-134 and Cs-137 reflects their chemical similarity and simultaneous release from the reactor. The significance of I-131 further supports the interconnected nature of radioactive isotope dispersion.

The small but significant effect of latitude indicates a geographical gradient in radiation dispersion, possibly influenced by atmospheric circulation patterns at the time of the accident. The insignificance of longitude suggests that contamination spread was not purely east-west but followed more complex environmental pathways.

Although the interaction model provides the best statistical fit, the base model is preferred for its clarity and ease of interpretation. The findings highlight the importance of using related radioactive isotopes as predictors in environmental radiation analysis.

Despite some violations of regression assumptions, such as non-normal residuals and heteroscedasticity, the model explains a large proportion of variance and provides meaningful insights. These limitations are common in real-world environmental data and do not invalidate the overall conclusions.

V. CONCLUSION

This study aimed to analyze and model the concentration of Cesium-137 (Cs-137) following the Chernobyl nuclear disaster using multiple linear regression techniques. By utilizing secondary data consisting of 1,415 observations across various European countries, this research investigated the relationships between Cs-137 and other radioactive isotopes, namely Cesium-134 (Cs-134) and Iodine-131 (I-

131), as well as geographic factors such as latitude and longitude.

The results consistently demonstrate that Cs-134 is the strongest predictor of Cs-137 concentrations. This finding is scientifically coherent, as both isotopes are cesium-based radionuclides released simultaneously during the reactor explosion. Iodine-131 also shows a statistically significant positive effect, confirming that regions with high iodine contamination tend to exhibit elevated cesium levels as well. These results highlight the interconnected nature of radioactive dispersion following a nuclear accident.

Geographic variables show weaker effects. Latitude exhibits a small but statistically significant negative relationship with Cs-137, suggesting a gradual decrease in contamination toward northern regions. Longitude, however, is not statistically significant in any model, indicating that east–west positioning alone does not adequately explain radiation variation once isotope concentrations are considered. This suggests that atmospheric conditions, rainfall patterns, and environmental deposition processes likely play a more dominant role than simple geographic coordinates.

Among the models tested, the interaction model achieved the highest statistical performance, indicating that the effect of Cs-134 on Cs-137 depends on the level of I-131. The fixed effects model further revealed that country-specific characteristics influence radiation levels, with several countries showing significantly different baseline concentrations. However, despite these improvements in fit, the basic linear regression model was selected as the final model due to its strong explanatory power, simplicity, and ease of interpretation. It explains nearly 79% of the variance in Cs-137 while remaining parsimonious and aligned with the primary objective of the study.

Diagnostic tests revealed violations of normality and homoscedasticity assumptions, which are common in environmental and radiation data due to skewed distributions and extreme values. Nevertheless, given the large sample size and strong statistical significance of key predictors, the model remains robust and suitable for academic interpretation.

In conclusion, this study confirms that Cs-134 and I-131 are the primary determinants of Cs-137 concentration in post-Chernobyl radiation data, while geographic location plays a secondary role. The findings demonstrate the effectiveness of multiple linear regression in modeling environmental

radiation data and provide valuable insights into radioactive dispersion patterns. Future research may extend this analysis by incorporating meteorological variables, time-series modeling, or spatial regression techniques to further improve predictive accuracy and capture complex dispersion dynamics.

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